

At The Speed Of Thought

Why can't computers read my mind?

Robert Sovereign-Smith

Have you ever wondered about the miracle that is the human body? Think about it: you go out, buy yourself this copy of *Digit*, sit down to read it, and have reached this article. Picking up this magazine, flipping pages, reading, understanding the words printed on this paper, forming an opinion about whether you like this article or not... all these actions seem to require absolutely no effort at all. Yet, in order to be able to accomplish this, the body has gone through hundreds of thousands of years of evolution, and is made up of one of the most complex networks (veins, arteries, nerves, muscles, food canal, etc...) known to mankind!

Every time you experience something, there's a system hard at work that's more complicated than all the computers of the world put together. Yet it just comes to us so naturally!

Now, think of the way you use a computer: you have to switch it on, wait for it to boot, then use a mouse and keyboard to try and get it to do what you want. There's complicated software to be handled, there's problems and limitations of the hardware, there are a myriad problems, but the worst of all is that you have to actually, physically, control input using motor co-ordination.

The Physicality Of It All

Now imagine if you could sit back, relax, and just look at your computer screen and then "think-type" instead of having to touch-type! What about if you could just focus on an icon and *think* "double-click" or "open" instead of actually doing so?

Sounds unbelievable? Well, the few lucky souls who visited the CeBIT exhibition in Germany know otherwise!

Called the Berlin Brain Computer Interface, the prototype on display had two people rigged up to a really weird-looking headset with a zillion wires attached to it, and then had them play the clas-

sic game *Pong*. You can see a video of it on YouTube.com (just search for "cebit thought"). Though it's not perfected yet, if you can move your bat in pong, in time, you should be able to do a lot more.

Your Thought Is My Command

Now here come the interesting bits: the beauty of thought computing doesn't just amount to being able to control your mouse with your thoughts, or silently telling your computer to launch Word, and then proceeding to *think* out that letter you were meaning to write...

Think (pun intended) about it: almost everything has a computer chip in it these days. From the washing machine to the toaster to your car, everything has a circuit board and some sort of firmware to control some function. It's not an over-imaginative mind that will foresee this sort of headset being made wireless—and hopefully a lot smaller as well!

Though it will take some time to achieve, we definitely seem to be on the right path towards thought controlling everything that can be controlled. We could see little chips being connected to our brain—permanently, like implants—and then be able to walk into a room and have the lights come on automatically, just by thinking, "It's dark in here!"

What about being able to drive a car just by sitting back and thinking about the direction you want to turn? Future wars will perhaps have no people actually fighting them, just thought-controlled machines attacking each other—may the stronger brain win!

The possibilities are endless, and as with every breakthrough technology, could be put to use in either useful or harmful ways. Of

course there are always concerns, especially when a technology interfaces so closely with our brains...

Controlled Thought?

Thought control is good, but controlled thought will have people up in arms! The fact is, what most of us would like is a one-way flow of information, from our brains to the machines. The reverse is not as pleasant. Though there could be a myriad of new inventions if machines could send information directly to our brains, something tells us that none of us would ever completely trust such a system.

Still, it could mean the end of display technologies: Imagine being able to experience a movie or game instead of just watching and listening to it?

For Now

We're just happy to finally be going to that next level of input devices, which sorely needed a revolution in order to make life easier. Even if it's just a way to ease our pain caused by carpal tunnel syndrome, because of excessive mouse and keyboard usage from cleaning up the spam from the hundreds of e-mails we receive each day, we'd rate thought-controlled computing as our deepest desire for the decade ahead! ■

robert_smith@thinkdigit.com

Illustration Pravin Warhokar

Resume